Huawei Access Node JAY-ACC-68 Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) Resolution SOP

Node: JAY-ACC-68 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_2917

Description

This standard operating procedure outlines the steps to diagnose and resolve an Optical Transport Failure on the Huawei Access Node JAY-ACC-68 located at the Jayanagar Metro Site. The failure domain in question is attributed to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. On Huawei equipment, the access node's optical transport functions are governed by the Optical Transport Network (OTN) protocol, which utilizes the SFP/XFP modules to transmit and receive optical signals. The OTN protocol operates at the physical layer, utilizing the Synchronous Optical Networking (SONET) or Synchronous Digital Hierarchy (SDH) framing formats to multiplex and demultiplex optical signals. In the event of SFP/XFP degradation or fiber cuts, the OTN protocol's ability to transmit and receive optical signals is compromised, resulting in a failure to establish or maintain connectivity. Furthermore, Rx/Tx power loss can also impact the OTN protocol's ability to function correctly, leading to a failure of the optical transport functions. This SOP outlines the steps to diagnose and resolve such failures on the Huawei Access Node JAY-ACC-68.

Pre-requisite Checks

- Ensure that the remote hands are dispatched to the Jayanagar Metro Site to gain physical access to the Huawei Access Node JAY-ACC-68.

- Verify that the redundant link status is functioning correctly and that the secondary link is not impacted by the failure.

- Power cycle the Huawei Access Node JAY-ACC-68 to ensure that all interfaces are reset and functioning correctly.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-transport interface, Expected Output:
Interface Admin Oper Type Tx Power Rx Power --------------- ---------
--------- --------- ------------ ------------ GE1/0/0 Up Down Ethernet
```

```
-28.56 -33.22, Description: This command displays the current state of the optical transport interfaces on the Huawei Access Node JAY-ACC-68. The expected output indicates that the GE1/0/0 interface is down, which may be indicative of a SFP/XFP degradation or fiber cut.

Command: display optical-transport sfp, Expected Output: SFP Admin Oper Type Tx Power Rx Power --------- --------- --------- --------- ------------ ------------ GE1/0/0 Down Down Ethernet -28.56 -33.22, Description: This command displays the current state of the SFP/XFP modules on the Huawei Access Node JAY-ACC-68. The expected output indicates that the GE1/0/0 SFP is down, which may be indicative of a SFP/XFP degradation.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace optical-transport interface GE1/0/0, Description: Replace the SFP/XFP module on the GE1/0/0 interface with a new one to resolve the SFP/XFP degradation issue.

Step 2: Step: Verify the fiber connectivity, Command: display optical-transport interface, Description: Verify the fiber connectivity by checking the status of the GE1/0/0 interface. If the interface is still down, it may indicate a fiber cut issue.

Post-Check Verification

```
Command: display optical-transport interface, Expected Output: Interface Admin Oper Type Tx Power Rx Power --------------- --------- --------- --------- ------------ ------------ GE1/0/0 Up Up Ethernet -28.56 -33.22, Description: This command displays the current state of the optical transport interfaces on the Huawei Access Node JAY-ACC-68. The expected output indicates that the GE1/0/0 interface is up and functioning correctly.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: undo replace optical-transport interface GE1/0/0, Description: Restore the original SFP/XFP module on the GE1/0/0 interface to resolve the issue in case the replacement module does not resolve the problem.